<table>
<tr><td>4(a)(i)</td><td>M1 (molecules with the) same molecular formula / same number of atoms of each element

M2 different structural formulae/structures</td><td>2</td></tr>
<tr><td>4(b)(i)</td><td>KCN in ethanol (and heat)</td><td>1</td></tr>
<tr><td>4(b)(iii)</td><td><table><tr><th></th><th>reaction of V with propan-2-ol</th></tr><tr><td>type of reaction</td><td>M1 condensation</td></tr><tr><td>functional group formed</td><td>M2 ester</td></tr><tr><td>molecular formula of organic product W</td><td>M3 $C_9H_{18}O_2$</td></tr></table></td><td>3</td></tr>
</table>

Examiner report missing
9701_w25_qp_24 Question 4

4(a)(i)	reduction	**1**

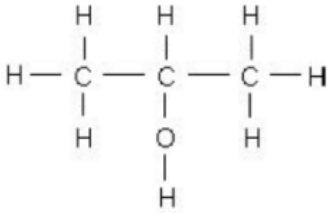
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9701_w25_qp_23 Question 4

4(a)(i)	free-radical substitution	**1**
4(d)(i)	condensation	**1**

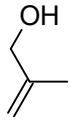
Examiner report missing
9701_w25_qp_22 Question 4

4(a)(i)	reduction	**1**

Examiner report missing
9701_w25_qp_21 Question 4

4(a)	$C_5H_{10}O_2$	**1**
4(b)	propyl ethanoate	**1**
4(c)(i)	condensation	**1**
4(c)(ii)	$CH_3COOH + CH_3CH_2CH_2OH \rightarrow CH_3COOCH_2CH_2CH_3 + H_2O$	**1**
4(c)(iii)	propan-1-ol	**1**
4(d)(i)	$CH_3CN + HCl + 2H_2O \rightarrow CH_3COOH + NH_4Cl$	**1**
4(d)(ii)		**2**

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9701_s25_qp_24 Question 4

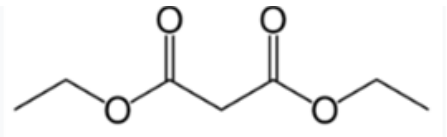
5(a)(i)	C_2H_3O	**1**
5(b)(ii)		**1**
5(b)(iv)	methanol / CH_3OH AND conc(entrated) (sulfuric) acid (catalyst)	**1**

Examiner report missing
9701_s25_qp_23 Question 5

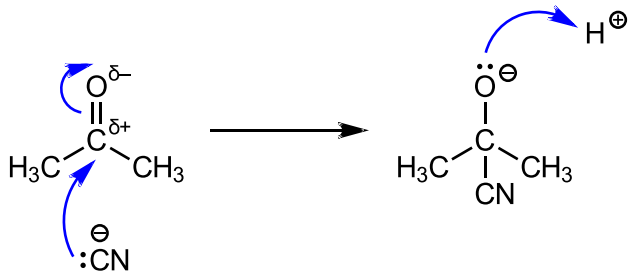
4(a)(i)	$LiAlH_4$	**1**

4(a)(ii)	$CH_3CH_2CO_2H + 4[H] \rightarrow CH_3CH_2CH_2OH + H_2O$	**1**

Examiner report missing
9701_s25_qp_22 Question 4

5(a)	**M1** two ester links **M2** rest of structure correct 	**2**
5(b)	(acid) hydrolysis	**1**
5(c)(i)	$HOOCCH_2COOH + Na_2CO_3 \rightarrow NaOOCCH_2COONa + CO_2 + H_2O$	**1**
5(e)(i)	$LiAlH_4$	**1**
5(e)(ii)	**M1** correct unambiguous organic product **M2** correct balanced equation $HOOCCH_2COOH + 8[H] \rightarrow HOCH_2CH_2CH_2OH + 2H_2O$	**2**

Examiner report missing
9701_s25_qp_21 Question 5

4(a)(i)	 • curly arrow from lone pair on C of CN^- to carbonyl C • correct dipole on C=O • curly arrow from C=O bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O to (source of) H^+	**1**
		1
		1

4(b)	H_2SO_4(aq) or HC*l*(aq)	**1**

Question 4

(a) (i) The mechanism attracted many good answers. The common errors included: arrows from the negative charge (on the CN^- ion) instead of from the lone pair on the C atom, missing polarity, specifically missing the $\delta+$ on the H (of H-CN), and incorrect attack of the intermediate by the CN^- ion.

(ii) Many answers stated that there were two CH_3 groups on one C atom in A but did not go further to explain the significance of this in terms of the lack of optical isomerism in A. Credit was given for stating that no chiral centre was present or that the central C atom was not bonded to four different atoms/groups.

(b) Candidates often stated that the mineral acids HCl or H_2SO_4 were required for the hydrolysis reaction of a CN group to a carboxylic acid but missed the requirement for the acid to be dilute or aqueous.

(c) (i) The equation was frequently correct. A common error was the omission of 2(H), from the equation, or using just 2H, without the brackets.

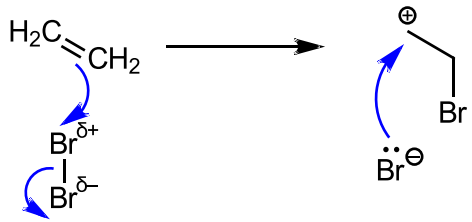
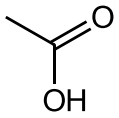
(ii) The name of compound C was well known.

(d) Most candidate answers stated that an orange precipitate was formed when propanone was reacted with 2,4-DNPH.

(e) A common answer was that 'Fehling's reagent only reacts with aldehydes', with no reference to any chemistry in this reaction. Fehling's reagent behaves as an oxidising reagent and ketones are not easily oxidised, hence, there is no reaction between Fehling's reagent and propanone.

(f) (i) Many candidate answers stated that the absorptions at 2850 – 2950 cm^{-1} were attributable to a C-H bond. As all three compounds in the question contained groups containing C-H bonds these compounds were indistinguishable using these absorptions.

(ii) Many candidates correctly identified the absorption at 2200 – 2250cm^{-1}, which was unique to compound A, as it was the only compound containing a C≡N bond. As all three compounds, A, B and C contained an O-H bond, no credit was given for frequent mention of this absorption.

4(a)	 **pt 1** dipole on Br_2 **pt 2** curly arrow from C=C bond to a $Br^{\delta+}$ **pt 3** curly arrow from Br–Br bond to the other Br **pt 4** correct intermediate **pt 5** curly arrow from lone pair on Br^- to C^+ **5•** = 3 marks. **4•** = 2 marks. **3•** = 2 marks. **2•** = 1 marks. **1•** = 0 marks.	**3**
4(d)(i)	structural / positional	**1**
4(e)(i)	**M1** identify **H** =  **M2** (broad) absorption within the range 3600–2500 cm^{-1} so O-H (bond) O-H (bond) is equivalent to OH bond **M3** pt 1 and pt 2 •✓ **OR** pt 1 and pt 3 •✓ •**pt 1** absorption within the range 1670–1750 cm^{-1} so C=O (bond) •**pt 2** absorption within the range 1040–1300 cm^{-1} so C-O (bond) •**pt 3** (M^+ at m/e = 60 so it has) molecular mass / Mr = 60	
4(e)(ii)	oxidising agent	**1**

Question 4

(a) Well-rehearsed candidate answers included the accurate placement of curly arrows, dipoles and the correct intermediate. Weaker answers often had ambiguous placement of curly arrows, inappropriate use of partial charges, incorrect dipole on Br_2 and incorrect intermediates.

(b) Correct calculations, sometimes with an appropriate Hess' cycle in their working, were seen. Incorrect answers were calculated by those candidates that did not fully understand or use the definition of enthalpy change of formation when applying Hess's law.

(c) (i) The product **A** was correctly shown in many answers. Candidates need to show correct connectivity of the O-H group to the carbon backbone when structures are drawn.
The naming of **B** presented problems for some candidates. Incorrect names often described the incorrect position of Br on the carbon backbone or gave a name which described the wrong number of carbon atoms in the backbone.

(ii) Unambiguous structures were seen in a significant number of candidate answers.

(iii) This question proved to be extremely demanding. All the information provided in the details of Fig. 5.1 needed to be considered, including the three products as well as the reaction conditions.

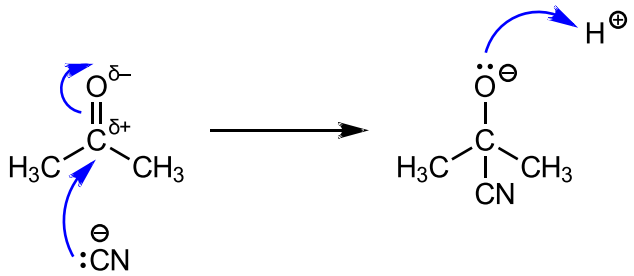
(d) (i) Correct identification of the type of isomerism was given in a significant number of candidate answers. Chain isomerism was a common incorrect answer even though the two molecules were made up of only two carbon atoms.

(ii) Many candidate answers stated that **F** was a carbonyl compound; this term describes the functional group present in the molecule and is common to both homologous series; aldehyde and ketone.

(iii) Many candidate answers gave appropriate observations for the behaviour of **F** with 2,4-DNPH reagent and Tollen's reagent. A much smaller number of answers identified a positive reaction with appropriate observation for the reaction with alkaline I_2(aq).

(e) (i) This question proved to be demanding. The strongest candidates gave full explanations of their interpretation of the infrared spectrum by relating relevant wavenumber values at absorption troughs to appropriate bond types and used the molecular ion peak to deduce the molecular mass of **H**. Weaker responses quoted key wavenumber values at troughs in the infrared spectrum and restated that the molecular ion peak in the mass spectrum was at m/e = 60 without explaining what this information actually provides about structure **H**.

(ii) The correct role of reagent **G** was given in many candidate answers. Catalyst was a common incorrect answer.

4(a)(i)	 • curly arrow from lone pair on C of CN^- to carbonyl C • correct dipole on C=O • curly arrow from C=O bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O to (source of) H^+	**1**
		1
		1

4(b)	H_2SO_4(aq) or HC*l*(aq)	**1**

Question 4

(a) (i) The mechanism attracted many good answers. The common errors included: arrows from the negative charge (on the CN^- ion) instead of from the lone pair on the C atom, missing polarity, specifically missing the $\delta+$ on the H (of H-CN), and incorrect attack of the intermediate by the CN^- ion.

(ii) Many answers stated that there were two CH_3 groups on one C atom in A but did not go further to explain the significance of this in terms of the lack of optical isomerism in A. Credit was given for stating that no chiral centre was present or that the central C atom was not bonded to four different atoms/groups.

(b) Candidates often stated that the mineral acids HCl or H_2SO_4 were required for the hydrolysis reaction of a CN group to a carboxylic acid but missed the requirement for the acid to be dilute or aqueous.

(c) (i) The equation was frequently correct. A common error was the omission of 2(H), from the equation, or using just 2H, without the brackets.

(ii) The name of compound C was well known.

(d) Most candidate answers stated that an orange precipitate was formed when propanone was reacted with 2,4-DNPH.

(e) A common answer was that 'Fehling's reagent only reacts with aldehydes', with no reference to any chemistry in this reaction. Fehling's reagent behaves as an oxidising reagent and ketones are not easily oxidised, hence, there is no reaction between Fehling's reagent and propanone.

(f) (i) Many candidate answers stated that the absorptions at 2850 – 2950 cm^{-1} were attributable to a C-H bond. As all three compounds in the question contained groups containing C-H bonds these compounds were indistinguishable using these absorptions.

(ii) Many candidates correctly identified the absorption at 2200 – 2250cm^{-1}, which was unique to compound A, as it was the only compound containing a C≡N bond. As all three compounds, A, B and C contained an O-H bond, no credit was given for frequent mention of this absorption.

5(a)	cold dilute $KMnO_4$ **OR** cold dilute potassium manganate(VII)	**1**
5(d)(i)	**W** =	**1**
5(d)(ii)	NH_4Cl	**1**
5(d)(iii)	$C_8H_{12}O_4 + 8[H] \rightarrow C_8H_{16}O_2 + 2H_2O$	**1**
5(d)(iv)	$LiAlH_4$	**1**

Question 5

This question tested organic reactions and spectroscopic methods using cyclohexene as a starting material. Many candidates seemed familiar with the chemistry, although the context of a cyclic molecule proved to be disconcerting for a few, who appeared to misinterpret the information given, or did not concentrate solely on the relevant functional groups.

(a) This was well answered by many, although differences appeared in the conditions to use with the $KMnO_4$ reagent (e.g. concentrated/dilute, alkaline/neutral/acidic).

(b)(i) Step 1 was familiar to many candidates. The hydrolysis of two functional groups in step 2 led many to think that H_2O_2 was being used, rather than two portions of NaOH.

(ii) In asking for a mechanism, 'electrophilic' was expected as well as 'addition'.

(c) Candidates showed some understanding in answering this question but often missed full credit by not following the rubric properly: the diagram needed labelling and each responsible bond (stretch) needed to be clearly identified. Many candidates only achieved half of this, or switched **F** and **G**.

(d)(i) This was mostly well answered, although some candidates lost credit when opting to give fully displayed versions of the CN group, neglecting to show a triple C≡N bond or using the wrong connectivity (N attached to the ring).

(ii) This was answered well by many candidates. Others did not acknowledge that acidic conditions are used, creating an ammonium species rather than NH_3.

(iii) The formulae of the organic products were generally well indicated, though candidates often then misbalanced the equation with insufficient (H).

(iv) This was generally well answered by candidates. A few $NaBH_4$ responses were offered, to no credit: carboxylic acids need the stronger of the two hydride reagents on the syllabus.

4(a)	<table><tr><td>molecule</td><td>**P**</td><td>**Q**</td><td>**R**</td><td>**S**</td></tr><tr><td>geometrical</td><td>✓</td><td>(×)</td><td>(×)</td><td>(×)</td></tr><tr><td>optical</td><td>(×)</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	**1**
	<table><tr><td>molecule</td><td>**P**</td><td>**Q**</td><td>**R**</td><td>**S**</td></tr><tr><td>geometrical</td><td>✓</td><td>(×)</td><td>(×)</td><td>(×)</td></tr><tr><td>optical</td><td>(×)</td><td>✓</td><td>✓</td><td>✓</td></tr></table> $CH_3CHBrCH_2CH_3$	**1**

4(c)(i)	n = 100 / 1.1 × 0.55 / 7.2	**1**
	= 6.94 (so 7 carbon atoms)	**1**

4(c)(ii)	FG = ester / (R)COOR(')	**1**
	(absorption at) 1040–1300 cm^{-1} indicates C—O (stretch) **AND** (absorption at) 1710–1750 cm^{-1} indicates (C=O stretch)	**1**

Question 4

This an organic and spectroscopic question. Candidates are advised to be clear in matters of structure, both when representing molecules diagrammatically and when referring to features within a molecule, such as those identified by different types of spectra.

(a) This was answered well by many candidates. There were incidences of ambiguous answers, such as a 'crossed tick' (✓✗). Credit cannot be awarded if an answer is ambiguous.

(b) (i) Unambiguous structural formulae only were accepted, therefore molecular formulae were discounted and those in which the connectivity was questionable.

(ii) There were many correct answers to this question.

(iii) This proved to be one of the most challenging questions on the paper, requiring the balanced inclusion of (2)H_2O and H^+ on the left-hand side, and NH_4^+ on the right. Salts from viable reactions of specific acids were accepted.

(c) (i) Candidates who understood the $M^+ : [M+1]^+$ ratio were generally able to gain full credit here. Occasional arithmetical errors were seen.

(ii) Many candidates gave 'ester' as a correct answer but were then unable to corroborate this with sufficient and necessary information from the spectrum. Reference to an absorption outside of the C=O ester region contradicted an answer; equally, an absorption within the 1710–1750 cm^{-1} window was not solely sufficient, because of overlap with other absorptions. Corroboration could be given with reference to the C–O stretch (and the lack of O–H stretch).

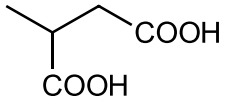
C–H stretches are always ignored in these situations.

Absorptions must be accompanied by with assignment to an identified bond.

(iii) Propan-1-ol was a common incorrect answer for **T**. OH– is discouraged as a representation of an alcohol group. Many candidates did not draw the branched C5 acid correctly: some structures were unbranched; others had too many or too few H atoms.

2(a)(i)	increases down the group	**1**

2(a)(iv)	$Ba(CH_3COO)_2$ / barium ethanoate	**1**

2(c)	CH_3OH	**1**
	 one COOH group **AND** correct carbon framework (5C's) (each C has the correct valency)	**1**
	both COOH groups	**1**

Question 2

(a) (i) The solubilities of group 2 hydroxides 'increases' down the group; this variation in solubilities was correctly stated by many candidates

(ii) When barium oxide is added to water the observation is that the white solid dissolves/disappears to form a colourless solution. A common error was that there was bubbling/effervescence seen.

(iii) In the given reaction scheme, reaction 2 involved the formation of $BaCO_3$ and water from adding a reactant to $Ba(OH)_2$. Carbon dioxide, CO_2, is a reactant that would form both the carbonate and water. Many answers included a metal carbonate which would not give water as a product.

(iv) The addition of ethanoic acid to $Ba(OH)_2$ gave barium ethanoate, $(CH_3COO)_2Ba$, as the product.

Many candidates appreciated that barium ethanoate was formed but opted to answer using a formula of $(CH_3COO)Ba$, which was not creditworthy.

(v) $Ba(OH)_2$ and hydrogen are formed when Ba is added to water and the equation is:

$$Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$$

A common error was to use only 1 mole of water molecules and water formed as a product.

(b) (i) The thermal stability of group 2 carbonates increases down the group; hence, when barytocalcite, $BaCa(CO_3)_2$, is heated $CaCO_3$ will decompose first. Most answers correctly stated that $CaCO_3$ would decompose first but explanations compared the reactivity of Ca with Ba.

(ii) The equation to represent the thermal decomposition of $BaCa(CO_3)_2$ was often correct.

$$BaCa(CO_3)_2 \rightarrow BaO + CaO + 2CO_2$$

Common errors included answers with BaCa as a product or omitting $2CO_2$ as a product.

(c) Full credit for the hydrolysis reaction was rarely awarded. The two organic products are methanol, CH_3OH, and the diacid.

That both the –CN and $–CO_2CH_3$ groups will undergo hydrolysis was not recognised by many candidates. The –CN group was frequently left unhydrolysed and the product methanol was not given as the second organic product.

3(a)(i)	$4P / P_4 + 5O_2 \rightarrow P_4O_{10}$	**1**
	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	**1**
3(c)(i)	H_2O / steam	**1**
3(c)(iii)	condensation	**1**
3(d)(i)	$H_3PO_4 + 3CH_3OH \rightarrow (CH_3O)_3PO + 3H_2O$	**1**

Question 3

(a) (i) Fully balanced equations were seen by more able candidates.

(ii) This definition was described accurately in most candidates answers. A common misconception was seen in weaker responses that described a weak acid as one that *'did not completely dissolve'*.

(b) (i) Many candidates gave a definition of the enthalpy change which occurs when 'one mole of product is formed from its elements'. It was less common for the elements to be described as 'in their standard states'.

(ii) More able candidates gave excellent and logical workings that contained clearly labelled energy cycles. It was common to see an expression for the different bond energies added or subtracted from each other, in a variety of possible incorrect combinations, with no attempt to show how these expressions were deduced. The stoichiometry of the equation was ignored by some.

(iii) Excellent answers were seen which identified phosphorus in H_3PO_3 as both oxidised and reduced and linked the process of oxidation and reduction to the relevant changes in oxidation number for the three phosphorus containing species. Weaker responses were less specific, for example '*H_3PO_3 is oxidised and reduced'*. Oxidation numbers were sometimes described incorrectly.

(c) (i) Some candidates identified **A** as steam or $H_2O(g)$, other candidates were less precise and identified **A** as water. A common incorrect answer was NaOH.

(ii) Many unambiguous representations of 2-bromopropane were seen. Occasionally the central carbon atom was drawn with a valency of 3.

(iii) Description of this esterification reaction as a condensation reaction was relatively well known. Incorrect answers included hydrolysis and oxidation.

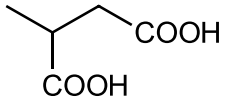
(iv) This question proved to be demanding. Candidates were expected to ignore all bonds which were common to both reactant and product. While the appropriate bond, C=O, was identified in many answers C-O and C-H were also seen. Most candidates only identified a range of wavenumbers where a change in absorption occurs rather than described if the absorption appeared, disappeared or moved.

(d) (i) More able candidates deduced a formula for the correct organic product in the equation. The inclusion of H_2O as the additional product was less common, even when included the equation was not always balanced.

(ii) The appropriate formula to deduce the number of carbon atoms present in **T** was not well known and many confused calculations were seen. It was common for candidates to give a structural formula for **T** rather than the required molecular formula.

2(a)(i)	increases down the group	**1**

2(a)(iv)	$Ba(CH_3COO)_2$ / barium ethanoate	**1**

2(c)	CH_3OH	**1**
	 one COOH group **AND** correct carbon framework (5C's) (each C has the correct valency)	**1**
	both COOH groups	**1**

Question 2

(a) (i) The solubilities of group 2 hydroxides 'increases' down the group; this variation in solubilities was correctly stated by many candidates

(ii) When barium oxide is added to water the observation is that the white solid dissolves/disappears to form a colourless solution. A common error was that there was bubbling/effervescence seen.

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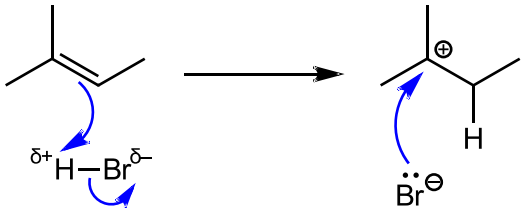
$$BaCa(CO_3)_2 \rightarrow BaO + CaO + 2CO_2$$

Common errors included answers with BaCa as a product or omitting $2CO_2$ as a product.

(c) Full credit for the hydrolysis reaction was rarely awarded. The two organic products are methanol, CH_3OH, and the diacid.

That both the –CN and $–CO_2CH_3$ groups will undergo hydrolysis was not recognised by many candidates. The –CN group was frequently left unhydrolysed and the product methanol was not given as the second organic product.

4(a)(i)	3-methylbut-1-ene	**1**

4(c)(i)	 correct dipole on H-Br AND curly arrow from C=C bond to $H^{\sigma+}$ **AND** curly arrow from H—Br bond to $Br^{\sigma-}$	**1**
	correct (Markovnikoff) intermediate	**1**
	curly arrow from lone pair on Br^- to C^+ of intermediate	**1**

4(d)	(reaction 2) NaOH**(aq)**	**1**
	(reaction 3) hot **acidified <u>concentrated</u>** $KMnO_4$	**1**
	(reaction 4) <u>conc</u> H_2SO_4 catalyst	**1**

Question 4

(a) (i) Most candidates correctly gave the systematic name of the alkene **D**, with a molecular formula of C_5H_{10}, as **3-methylbut-1-ene**.

(ii) Few candidates gave a correct explanation of why the alkenes, **C** and **D**, did not exhibit geometric isomerism.

In **C**, one C (in the C=C bond) has two methyl groups and in **D**, one C (in the C=C) has two hydrogen atoms; neither will exhibit geometrical isomerism.

(iii) Many candidates gave pent-2-ene, or **C** or **D** as examples of a positional isomer.

(iv) When **D** is reacted with $H_2(g)$, 2-methylbutane, $CH_3CH(CH_3)CH_2CH_3$, is formed. Many candidates omitted the brackets surrounding the side-chain methyl group, on the CH group.

(v) Many candidates gave a monomer containing only two methyl groups or structures that still contained a double bond.

(b) The question contained information regarding the relative intensities of the M and (M + 1) ions, from a mass spectrum of **C**, which showed a molecular ion peak at 70.

Many candidates correctly calculated that C contained 5C atoms but then failed to use this information, in conjunction with the molecular ion peak, 70, to confirm that the molecular formula of C was C_5H_{10}.

(c) (i) The mechanism for the addition of HBr to **C,** 2-methylbut-2-ene, was quite well answered. Common errors included full charges on the H-Br, instead of a dipole, curly arrows, from C=C to $H^{\delta+}$ of HBr, arrows starting at one end of the C=C bond and intermediates with both a –ve charge and a partial negative charge.

(ii) This question was well answered with a common error being the use of C_3H_7, instead of the unambiguous expanded version, $(CH_3)_2CH–$.

(iii) This question required an explanation as to why different major products were formed when HBr reacts separately with C and D.

Many candidates appreciated that methyl/alkyl groups are responsible for a positive inductive effect and that the more alkyl groups, surrounding the carbocation, the more stable is the carbocation intermediate. However, the application of this concept to the intermediates of C and D was not well recognised.

Common errors were statements that the tertiary halogenoalkane was more stable, in C, and the secondary halogenoalkane was more stable, in D, rather than their respective intermediate carbocations.

Many candidates quoted Markovnikov's rule without any further explanation.

(d) *Please note that due to an issue with diagram labelling in this question, full marks have been awarded to all candidates to make sure that no candidates were disadvantaged. This has been corrected in the published version of the paper.*

The question required completion of a table of reagents and conditions for a reaction scheme.

The formation of alcohol, **F**, from a halogenoalkane, **E**, required the use of NaOH(aq).

Cleavage of the double bond, in **C**, to form a carboxylic acid, **G**, and propanone, was carried out using hot acidified concentrated $KMnO_4$.

The formation of ester, **H**, from **F** and **G**, needed concentrated H_2SO_4 (as a catalyst).

3(a)(i)	KCN in ethanol + heat	**1**
3(a)(iv)	$2\ CH_3CH_2COOH + Ca \rightarrow (CH_3CH_2COO)_2Ca + H_2$ **M1** H_2 as product **M2** balanced equation based on correct formula of Y	**2**
3(b)	acid (conditions) / $H^+(aq)$ **AND** heat under reflux	**1**

Question 3

(a) (i) Some candidates incorrectly described the reagent and conditions required to convert carbonyl compounds to hydroxynitriles, rather than halogenoalkanes to nitriles.

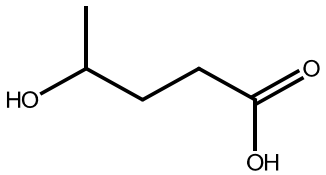
(ii) The systematic name for **X** was not always used, e.g. propionitrile or ethyl cyanide. Omission of the 'e' in propanenitrile was often seen.

(iii) It was generally well known that this type of reaction is a hydrolysis reaction.

(iv) Many candidates correctly produced equations representing the production of hydrogen gas. The formula of the calcium salt was frequently given in terms of calcium with a valency of one.

(b) Many candidates knew that acidic conditions were required. Stronger candidates appreciated that the essential condition for this reaction was 'heat under reflux' to ensure complete oxidation to the carboxylic acid.

(c) Most candidates described the correct number of pi bonds present in **X**; the correct number of sigma bonds was seen less frequently. Seven, nine and ten sigma bonds were common incorrect answers.

5(a)	molecules / isomers with the same molecular formula / same number of atoms of each element **AND** different structural formulae / different structures	**1**
5(d)(i)	ester	**1**
5(d)(ii)	condensation	**1**
5(d)(iii)		**1**

Question 5

(a) This definition was generally well known.

(b) (i) Many candidates appreciated that the two molecules were members of the carbonyl functional group. Some candidates incorrectly referred to them as either aldehydes or ketones.

(ii) Correct structures of **A** were common. Many candidates represented **B** as a ketone; the correct carbon skeleton was not always deduced.

(c) (i) It was common for some of the information in the question to be used to produce partially correct structures. Many candidates incorrectly produced structures with two types of stereoisomerism.

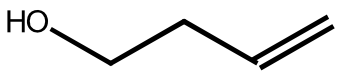
(ii) This was generally well answered.

(d) (i) Ketone and carboxylic acid were the most common incorrect answers.

(ii) Hydrolysis was a common incorrect response.

(iii) Deduction of the structure of **D** proved to be demanding. Often the structure drawn had a carboxylic acid group at one end of the molecule; identification of $(CH_3)CH(OH)$– at one end of the molecule was seen less frequently. Occasionally, the correct ends of the molecule were drawn with an incorrect number of carbon atoms in the carbon backbone.

(iv) Stronger candidates clearly identified two differences between the spectra in terms of the different absorption ranges produced as a result of the different environment of the C=O bond in an ester compared to a carboxyl group, or in terms of the presence of the O-H bond in a hydroxyl or carboxyl group which is absent in the ester, **E**. Reference to absorbance at 2850 – 2950 due to C-H bonds was common, even though both **D** and **E** contain this type of bond.

4(a)(i)		**1**

4(a)(ii)	reducing agent	**1**

Question 4

(a) (i) Common errors were to retain the COOH group and to have the C=C double bond reduced.

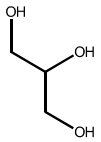
(ii) This was well answered by most candidates.

(b) (i) This was generally well answered, although few candidates stated all the required conditions.

(ii) This was well answered by most candidates.

(iii) A small proportion of candidates gained full credit here.

(c) Many candidates correctly gave pentan-1-ol. A small number of candidates made the link between the yellow precipitate with alkaline I_2(aq) and a $CH_3CH(OH)$– group. Many candidates struggled to synthesise all the detail to present a cogent answer to the question.

5(a)(i)	$C_8H_{18} + 12½\ O_2 \rightarrow 8CO_2 + 9H_2O$ **M1** correct formula of M C_8H_{18} **M2** balanced equation showing complete combustion of M	**2**
5(c)(i)	*suggest suitable **reagent** and conditions for step 1* heat **AND** any dilute named mineral acid e.g. dilute sulfuric acid / dilute hydrochloric acid	**1**
5(c)(ii)	 **OR** $CH_2OHCH(OH)CH_2OH$	**1**
5(c)(iii)	condensation	**1**
5(c)(iv)	methanol	**1**

Question 5

(a) (i) This was well answered by most candidates, although balancing the equation proved challenging for some.

(ii) Most candidates correctly mentioned the production of CO instead of CO_2. A very small number correctly referred to C and unburnt hydrocarbons.

(b) (i) A large proportion of candidates incorrectly interpreted this as a question about catalytic converters. Many answers used NO instead of N_2.

(ii) A small number of candidates correctly identified the catalytic role of NO_x and gave all the relevant equations. Many gave the equation for the formation of H_2SO_4, and a significant number gave equations and answers related to the formation of HNO_3 for acid rain.

(iii) This was well answered by most. Some candidates gave the identity of a gas instead of the type of air pollution.

(c) (i) Candidates often omitted dilute/(aq) with a named acid here.

(ii) A few candidates recognised that all three ester groups will undergo hydrolysis to form propanetriol. Many answers gave fragments where only one ester group had undergone hydrolysis.

(iii) This was well answered by most candidates. The majority of answers referred to esterification/substitution instead of condensation.

(iv) A small proportion of candidates answered this question correctly.

(v) A small proportion of candidates answered this question correctly.

4(a)(i)	NaOH in alcohol / ethanol AND heat (under reflux)	**1**

4(c)(i)	substitution	**1**

4(c)(ii)	reducing agent	**1**

Question 4

This was a second organic chemistry question, which included some analytical work. This was generally answered well by many candidates, though later parts on IR and MS appeared less familiar to some.

(a) (i) 'Heat' or equivalent was often missing. H_2SO_4 was a popular answer despite the conditions suggested in the syllabus.

(ii) Many repeat units were shown containing four carbon atoms, as if the CH_3 groups had been incorporated into the backbone. Some candidates showed an ethene structure with missing CH_3 side groups.

(iii) Candidates very often showed the reverse equation with C_4H_8 and H_2O. Quite a few answers contained Al_2O_3 as a reactant and several answers omitted H_2O as a product.

(b) (i) This was well answered.

(ii) There were many correct structures/names for iodoform but very few correct propanoate ions.

(c) (i) There were many correct answers to this item.

(ii) Catalyst and oxidising agent were the common errors here, but otherwise there were many correct answers.

(d) (i) Compound **F** was commonly given incorrectly as the unknown compound, referring to an O–H absorption in the range 2850–3000 cm^{-1}. Candidates are reminded that C–H stretches appear in all organic spectra. Where absorptions are referenced, candidates need to identify which bond is stretching: 'nitrile', for example, is insufficient.

(ii) There were many null responses to this. Where attempted, candidates' answers were evenly split between correct identification of an identical abundance and some complicated (erroneous) mathematics.